

# Sourdough at Sea — The Wild-Yeast Loaf

A living culture, a long cold proof, a hot covered bake. The wild loaf made reliable on a moving galley — and the acidity that flavours it also makes it keep.

Yield: **2 loaves** (10–12)    Active: **40 min**    Total: **2–3 days**    Oven: **250–230°C** (covered)    Done: **96–99°C** internal

|                                                                                                                                                                                                                                                                           |                                                                                                                                                                                         |                                                                                                                                                                                                               |
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| <b>WHERE IT WAS BORN</b><br>The oldest leaven there is — wild yeast in flour and water, the <i>only</i> way to raise bread before industrial yeast in the 1860s. Egyptian loaves >3,000 yrs ago; the name from Klondike prospectors who guarded the crock through winter. | <b>EATEN, AUTHENTICALLY</b><br>The everyday country loaf of every grain culture — torn warm with oil or butter; San Francisco gave it a famous tang and a bacterium named for the city. | <b>HOW IT’S USED NOW</b><br>The craft-bakery boule — and the chef’s living culture, carried boat to boat down the dock. Crumb for brunch, toast, bruschetta; stale, it makes <i>panzanella</i> and ribollita. |
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## INGREDIENTS

2 LOAVES · WEIGHT

### LEVAIN (NIGHT BEFORE)

|                                     |                |
|-------------------------------------|----------------|
| Mature starter ( <i>ripe</i> )      | 20g<br>0.71 oz |
| Bread flour                         | 100g<br>¾ cup  |
| Water ( <i>bottled / dechlor.</i> ) | 100g<br>⅔ cup  |

### FINAL DOUGH

|                                    |                 |
|------------------------------------|-----------------|
| Bread flour ( <i>12–13%</i> )      | 800g<br>6⅔ cups |
| Wholemeal / rye                    | 100g<br>¾ cup   |
| Water, cool ( <i>autolyse</i> )    | 600g<br>2½ cups |
| Water ( <i>reserved, w/ salt</i> ) | 50g<br>3⅓ tbsp  |
| Fine sea salt                      | 20g<br>1⅓ tbsp  |
| Levain ( <i>all, at peak</i> )     | ~200g<br>7.1 oz |

### SHAPING & BAKING

|                                  |               |
|----------------------------------|---------------|
| Rice flour ( <i>banneton</i> )   | 30g<br>3 tbsp |
| Semolina / flour ( <i>dust</i> ) | —             |

**THE RATIO — FLOUR = 100%**  
**Water 75% · Levain 20% · Salt 2%.** Four numbers; everything above is one batch of them, scaled for two crew loaves. Learn them and you never need the recipe again.

**Kit:** Dutch oven or combi/Rational · 2 bannetons (or towel-lined bowls) · lame / razor · scale · bench scraper · mixing tub · probe thermometer

## METHOD

1. **Build the levain.** Night before: 20g starter + 100g flour + 100g water. 4–8 h until doubled and a spoonful **floats**.

*A purpose-built levain gives a known, vigorous population at a known time; the float test reads enough gas to lift the loaf. — Robertson; Modernist Bread*

2. **Autolyse.** Mix 800g bread + 100g wholemeal flour with 600g water to a shaggy mass — **no levain, no salt**. Rest 45 min.

*Flour hydrates and gluten knits with no work; salt and acid now would tighten it too early. — McGee; López-Alt*

3. **Add levain, then salt.** Pinch in the ~200g levain; dissolve 20g salt in the reserved 50g water and work through.

*Dissolved salt spreads evenly and reins the yeast to a pace the long proof can ride. — Forkish*

4. **Bulk + 4 folds.** 24–26°C, ~4–5 h. Four stretch-and-folds in the first 2 h. Stop at ~40% risen, domed, jiggly.

*Folds build aligned gluten that traps big bubbles — the open crumb. Judge by rise, not the clock; a warm galley halves it. — Robertson*

5. **Pre-shape & shape.** Cut in two, pre-shape, bench 25 min, shape taut boules, **seam-up** in rice-floured bannetons.

*The rest stops tearing; a tight skin holds the loaf through proof and bake. Rice flour has no gluten, so it won’t glue. — Forkish*

6. **Cold proof 12–36 h.** Bag and into the fridge at 4°C. Bake straight from cold, no warm-up.

*Yeast crawls while bacteria build acid — deeper flavour, blistered crust, clean scoring; and it uncouples bake from proof so you fire to order. — Modernist Bread*

**Bake:** preheat Dutch oven/combi **250°C**. Turn out cold, **score** one slash, **lid on 20 min**; lid off, **230°C**, 20–25 min to deep mahogany / **96–99°C** internal. Cool **≥2 h** before cutting — the crumb sets as it cools.

**Keep it alive at sea** — the catch is the water. Chlorine, chloramine and stripped water-maker product stall a culture — feed bottled spring water, or stand tap water uncovered overnight. Warm galley (26–30°C) ferments fast (feed more, use cooler water); a cold fridge holds it dormant a week+ between feeds.

### CARRY IT ANYWHERE — DEHYDRATE

Smear ripe starter **paper-thin** on parchment, dry **below 40°C** (dehydrator, low oven, shaded deck), flake and **vacuum-seal**. Survives months and flights in a pocket; revive over two or three feeds. *Insurance against a crossing crew who bins your jar.*

ELEVATION

TIER 1 — NO EXTRA TIME

- Stiff levain (60% water) — milder, sweeter
- Longer cold proof — more tang, to 36 h
- Tin + foil lid — when shaping or sea won't cooperate

TIER 2 — WORTH IT

- Country loaf — 20–30% wholemeal/rye, +water
- Seeded — 150g soaked seeds after the salt
- 80–85% hydration — wilder, holier crumb

TIER 3 — SERVICE

- Olive & rosemary laminated in the folds
- Bâtard — more crust, easy sandwiches
- Discard → crackers, crumpets, blinis

| TROUBLESHOOT     |                                                                         |
|------------------|-------------------------------------------------------------------------|
| Dense, tight     | Levain not ripe / under-proofed → float test; bulk to ~40% domed        |
| Flat, spreads    | Over-proofed / weak shape → catch bulk earlier, shape taut, proof <36 h |
| Gummy centre     | Pulled pale → bake to 96–99°C; cool 2 h before cutting                  |
| Too sour         | Levain old / fridge warm → use younger, stiffer levain; verify 4°C      |
| Pale, no spring  | Pot/oven cool → true 250°C, deeper score, lid on 20 min                 |
| Starter sluggish | Chlorine / water-maker water → bottled water, warm corner               |

| CHARTER PREP & STORAGE |                                                                                 |
|------------------------|---------------------------------------------------------------------------------|
| Bake to order          | Hold shaped loaves 36 h at 4°C; fire from cold as the table needs               |
| Par-bake               | Covered to pale (~25 min); finish 10–12 min at 230°C at service                 |
| Frozen baked           | 1 mo wrapped; refresh 10 min at 180°C from frozen — freezes better than yeasted |
| Room temp              | Cut-face down, in cloth, 3–4 days — never the fridge (stales fastest)           |
| Carry the culture      | Dehydrate & vacuum-seal — months, flights, pocket-sized                         |